

Bandon Public Transport Accessibility Analysis

Bandon, County Cork, Republic of Ireland

SEL-002 | Public Transport Accessibility and Vulnerability
Prepared as a reproducible GIS-based spatial evidence study

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Research Questions and Objectives

Question 1: How does the spatial accessibility of public transport for travel between Bandon and Cork City vary across the town?

Question 2: Which areas are potentially most vulnerable to limited public transport accessibility because of their demographic, health, employment, or household characteristics?

Objectives

- 1. Map the spatial distribution of principal public transport stops and the surrounding network context.
- 2. Measure network-based walking accessibility to principal public transport stops.
- 3. Assess scheduled peak-period public transport service availability using GTFS-derived frequency and headway measures.
- 4. Construct a Population Vulnerability Index using selected Census 2022 indicators.
- 5. Compare spatial patterns of transport accessibility and population vulnerability.
- 6. Combine the resulting accessibility deficit and population vulnerability measures to identify spatial accessibility–vulnerability hotspots.
- 7. Identify locations that warrant further site-level or operational investigation.

Executive Summary

The analysis establishes a spatial screening framework across 34 Census Small Areas intersecting the Bandon Built-Up Area (BUA study population: 8,196 residents; Small Area aggregate population: 9,863). Transport accessibility is strongly centralised around the town core, with 64.7% (22 Small Areas) requiring network walking trips exceeding 10 minutes to reach principal stops. By decoupling physical network performance from demographic need prior to final synthesis, the model isolates 4 critical hotspot Small Areas where high relative population vulnerability ($VUL_norm > 0.75$) directly coincides with severe accessibility deficits (Access Deficit > 0.75), generating Hotspot Scores between 0.856 and 0.947.

Answers to the Research Questions

Research Question 1

How does the spatial accessibility of public transport for travel between Bandon and Cork City vary across the town?

Answer: Public transport accessibility is spatially differentiated across Bandon. Areas within and around the central accessibility corridor generally have shorter network walking journeys to principal stops and stronger scheduled peak-period service availability. Several peripheral residential areas have longer walking journeys and/or weaker scheduled service availability.

The analysis therefore identifies spatial differences in the conditions under which residents can access the Bandon–Cork public transport network. It does not establish whether public transport is universally viable or competitive with private-car travel because comparative journey times, travel costs, reliability, parking and individual mode choice are not modelled.

Research Question 2

Which areas are potentially most vulnerable to limited public transport accessibility because of their demographic, health, employment, or household characteristics?

Answer: The final hotspot analysis identifies areas where relatively weaker accessibility overlaps with higher relative population vulnerability. These locations occur primarily in peripheral residential areas. The vulnerability model incorporates older age, health and disability indicators, household car availability and selected economic-status indicators.

These classifications identify areas where limited accessibility may have greater planning significance. They should not be interpreted as direct measurements of individual transport dependence, individual disadvantage or observed unmet transport demand.

1. Understanding Public Transport Accessibility

A location may be physically close to a bus stop but still have relatively weak effective accessibility if services are infrequent. Conversely, a somewhat longer walk may have stronger effective accessibility where frequent services are available.

- Accessibility links transport provision to access to employment, education, healthcare, services and other opportunities.
- Mobility concerns movement itself; accessibility concerns the opportunities that movement enables.
- Transport equity requires consideration of both service provision and population need.

1.1 Public Transport Accessibility in Urban and Regional Planning

Accessibility is concerned with the relationship between people, transport systems and opportunities. A location may be physically close to a bus stop but still have relatively weak effective accessibility if services are infrequent. Conversely, a location with a somewhat longer walk may have stronger effective accessibility where frequent services are available. The Bandon analysis therefore separates walking access from service availability before combining them in the composite analysis.

- **Opportunities:** Accessibility links transport provision to access to employment, education, healthcare, services and other opportunities.
- **Mobility vs. Accessibility:** Mobility concerns movement itself, whereas accessibility concerns the opportunities that movement enables.
- **Quality of Life:** Accessibility is relevant to quality of life because transport conditions influence people's ability to participate in social and economic life.
- **Equity:** Transport equity requires consideration of both the distribution of service provision and the distribution of population need.

1.2 Accessibility as a Spatial Concept

Dimension	What the Bandon framework measures
Distance	How far a residential area is from a public transport stop through the pedestrian network.
Time	Estimated walking time required to reach the transport network.
Opportunity	Potential to use public transport once a stop is reached.
Network connectivity	Pedestrian routes available for reaching stops rather than straight-line distance.
Service availability	Frequency and temporal intensity of services derived from GTFS feeds.

INTERPRETATION

The Bandon analysis separates walking access from service availability before combining them in the composite analysis.

1.3 Accessibility in Small Towns

SPATIAL PATTERN

Smaller commuter towns can display a sharp transition between a relatively well-served urban core and lower-density peripheral residential areas.

- Bandon provides a case study in which accessibility is spatially structured rather than uniform across the study boundary.
- The town's central transport corridor generally has stronger network walking accessibility and scheduled service availability.
- Peripheral residential neighbourhoods can experience longer network walking journeys and/or weaker scheduled service availability.

Analytical distinction	Meaning in this study
Transport-system accessibility	Measured through network walking access and scheduled service availability.
Population vulnerability	Spatial concentration of selected Census characteristics associated with potential exposure to transport disadvantage.
Hotspot	Spatial overlap between access deficit and normalised population vulnerability.

1.4 Accessibility and Social Equity

IMPORTANT DISTINCTION

The vulnerability layer is not a measure of transport accessibility itself. It is a separate population-need dimension combined with accessibility only at the hotspot stage.

- Older age, health and disability, economic status and household car availability are treated as spatial proxies of potential vulnerability.
- These indicators describe relative Small Area conditions rather than individual-level characteristics.
- The final hotspot analysis identifies locations where relatively weaker accessibility overlaps with higher relative vulnerability.

Vulnerability is complementary evidence — not a direct measure of individual transport dependence.

1.5 Research Context

The Bandon study forms part of the Spatial Evidence Lab portfolio and applies a reproducible GIS approach to a small-town transport question.

CONCEPTUAL CHAIN

Transport Accessibility → Potential Mobility → Access to Opportunities → Potential Quality-of-Life Effects

2. Study Area and Analytical Framework

STUDY AREA

The analytical study area is defined using Census 2022 Built Up Areas. The boundary was used to select Small Areas included in the population-vulnerability analysis and to define the spatial extent of the transport-accessibility assessment.

- Results are normalised within the defined analytical population and should be interpreted relative to the Bandon study area.
- Bandon functions as a regional settlement and commuter node connecting to Cork City and surrounding West Cork settlements.
- The study focuses on the main settlement and surrounding residential areas represented within the defined analytical boundary.

2.1 Study Area Overview

The analysis distinguishes between the CSO Small Area geography used to construct the indicator dataset and the Bandon Built-Up Area (BUA) study boundary used to define the population and spatial extent of the study. This distinction is important because Small Areas do not necessarily align exactly with the Bandon BUA boundary. Some Small Areas intersect the study boundary only partially and therefore extend beyond the area included in the analysis.

The indicator dataset comprises 34 CSO Small Areas associated with the Bandon study area. The combined Census population of these 34 Small Areas is 9,863 persons. This figure represents the population of the complete Small Areas and should therefore not be interpreted as the population of the Bandon BUA itself.

For the purposes of this study, the principal study population is 8,196 residents, representing the population located within the defined Bandon BUA study boundary. The difference between the two figures arises from the partial spatial intersection between the Bandon BUA boundary and the CSO Small Area boundaries. In other words, the Small Areas provide the statistical geography for the indicator construction, while the Bandon BUA boundary defines the spatial extent of the study.

Accordingly, 8,196 residents is used as the headline population for the Bandon study, while the 9,863 figure is retained as the aggregate Census population of the 34 source Small Areas. The two figures should not be directly compared as alternative estimates of Bandon's population because they refer to different spatial units.

This distinction also establishes an important methodological limitation: Small Area-based demographic indicators are calculated using the available Census geography, whereas the study boundary is defined independently at BUA level. Consequently, indicators derived directly from complete Small Area populations should be interpreted as Small Area-level contextual measures, rather than as estimates of the characteristics of only the population falling inside the Bandon BUA boundary.

2.2 Study Context and Population Vulnerability Assessment

Bandon has a resident population of 8,196 (Census 2022) and functions as a regional settlement and commuter node. Bandon was selected because its defined urban core and surrounding lower-density residential areas provide a suitable case study for examining spatial variation in public transport accessibility.

Population Vulnerability as a Complementary Dimension: To examine the social context of transport accessibility, the analysis incorporates Census 2022 Small Area data to construct a Population Vulnerability Index. This index does not measure public transport accessibility or observed transport demand. Instead, it identifies spatial concentrations of selected demographic, health, economic-status and household characteristics that

may indicate greater potential exposure to transport disadvantage. The vulnerability index is therefore treated as a complementary population-need dimension. It is combined with the independently calculated accessibility measures only at the final hotspot stage.

Dimensions of Population Vulnerability

Census Variable	Description	Relevance to Vulnerability Assessment
T1_1AGE70_74T	Population aged 70-74	Identifies older residents who may have increased dependence on accessible transport.
T1_1AGE75_79T	Population aged 75-79	Represents populations with potentially greater mobility needs.
T1_1AGE80_84T	Population aged 80-84	Captures areas with higher concentrations of elderly residents.
T1_1AGEGE_85T	Population aged 85+	Represents population most likely to require highly accessible services.
T1_1AGETT	Total population	Denominator for population-based vulnerability measures.
T4_5RP / T8_1_RT	Retired persons	Represents populations less likely to rely on work travel but dependent on local transport.
T8_1_LTUT	Long-term unemployed	Represents socio-economic vulnerability and transport disadvantage.
T8_1_UTWSDT	Permanent sickness/disability	Represents populations experiencing greater mobility constraints.
T12_1_T	Persons with a disability	Measures populations requiring accessible transport infrastructure.
T12_3_BT	Bad general health	Indicates communities with health-related transport needs.
T12_3_VBT	Very bad general health	Identifies areas with high health-related transport vulnerability.
T15_1_NC	Households without a car	Households potentially most dependent on public transport.
T11_1_BUT	Travelling by bus/minibus	Provides evidence of existing public transport use.
T11_2_T1-T8	Departure time for work/school	Captures commuting demand during morning departure periods.
T11_2_NS / T11_2_T	Leaving time not stated / Total	Denominator for analysing commuting time distributions.

The selected Census categories may overlap at individual level; composite measures therefore represent model-derived concentrations rather than counts of unique vulnerable residents.

The selected Census variables are grouped into four analytical domains:

- **1. Demographic Vulnerability:** Older population, particularly residents aged 70 years and over.
- **2. Health and Mobility Vulnerability:** Disability, permanent sickness, and bad or very bad general health.
- **3. Socio-Economic Vulnerability:** Long-term unemployment and selected economic-status characteristics associated with potential transport disadvantage.
- **4. Household Transport Availability:** Households without access to a motor car.

Additional Census variables relating to bus use and journey departure times are retained as contextual evidence of observed travel behaviour and commuting patterns. They are not incorporated directly into the Population Vulnerability formula.

Integration Framework

The analysis strictly separates the spatial performance of the transport network from population demand and vulnerability:

Census Variable Domain	Purpose in the Study
Public Transport Supply and Accessibility	Measured via bus stop locations, walking-network catchments, and peak service frequency
Observed Travel Behaviour	Represented by Census commuting patterns and bus-use variable
Population Vulnerability	Represented by age, retirement, unemployment, disability, and health indicators
Private Vehicle Availability	Represented by households without access to a motor car

2.3 Analytical Framework and Map Series

Map	Title	Subtitle
01.00	Base Context & Transit Infrastructure	Bandon Built-Up Areas (BUA) and Network Baseline
01.01	Physical Pedestrian Catchments	Network-Based Isochronal Walkability Analysis
01.02	Pedestrian Network Context	Footpath Coverage and Pedestrian Crossing Networks
01.03	Headway & Operational Frequency	Peak-Period Transit Service Quality
01.04	Demographic Vulnerability & Demand	Spatial Distribution of Relative Population Vulnerability
01.05	Composite Accessibility Index (CAI)	Combined Pedestrian Accessibility and Scheduled Service Availability
01.06	Targeted Intervention Framework	Spatial Screening Priorities for Further Investigation

3. Dimensions and Methodology

3.1 Terminology and Model Architecture

To ensure consistency throughout the report, GIS databases, cartographic products, and web dashboards, the standardised terminology presented below must be strictly adhered to across all analytical workflows.

The vulnerability indicators represent spatial concentrations of selected Census characteristics and should not be interpreted as direct measures of individual transport dependence, transport need or individual disadvantage. Unless explicitly stated otherwise, the results describe relative Small Area conditions rather than individual-level characteristics.

Concept / Measure	Standardised Terminology	Application Scope and Context
Network walking access score	Pedestrian Accessibility	ArcGIS Network Analyst walking-access score based on network travel-time bands.
Combined transport accessibility score	Composite Accessibility Index (CAI)	50:50 combination of Pedestrian Accessibility and normalised scheduled service availability.
Pedestrian infrastructure contextual layer	Pedestrian Network Context	Footpaths, crossings and pedestrian-network features used to contextualise network accessibility; not independently weighted in the CAI.
Inverse CAI measure	Access Deficit	$1 - \text{CAI_norm}$; measures relative accessibility deficit.
Continuous spatial priority	Hotspot Score	40% Access Deficit + 60% Normalised Population Vulnerability.
Planning classification	Intervention Priority	Five-tier spatial screening classification derived from the Hotspot Score.

METHODOLOGICAL GUARDRAIL: INTERPRETATION OF SMALL AREA INDICATORS

The Population Vulnerability Index utilises min-max normalised Census 2022 indicators across four weighted domains: Elderly Population (30%), Health and Disability (30%), Household Car Availability (20%), and Economic Vulnerability (20%). These composite measures reflect spatial concentrations of demographic characteristics at the Small Area level rather than individual-level counts of transport-dependent residents. Pedestrian accessibility scores derived from ArcGIS Network Analyst walking bands (0-5 min = 1, 5-10 min = 0.6, 10-15 min = 0.3 and >15 min = 0) are reported as continuous network accessibility indices rather than simple radial percentages.

3.2 Analytical Workflow

The analytical pipeline separates transport-system accessibility from population vulnerability before combining the two dimensions at the final hotspot stage. Pedestrian Accessibility and scheduled Service Availability are first combined to produce the Composite Accessibility Index (CAI). Population Vulnerability is calculated independently of Census indicators and is not included in the CAI. The resulting CAI is normalised and converted into an Access Deficit. The Access Deficit is then combined with normalised Population Vulnerability to produce the final Hotspot Score.

ANALYTICAL WORKFLOW PIPELINE

Separate transport accessibility and population vulnerability before final spatial screening.

- Step 1: Census 2022 indicators extracted
- Step 2: Vulnerability index constructed (30/30/20/20 weights)
- Step 3: GTFS service frequency and headway evaluated
- Step 4: Network Analyst walking catchments calculated
- Step 5: CAI calculated (50:50 Pedestrian + Service)
- Step 6: Hotspot Score generated (40% Access Deficit + 60% VUL_norm)

3.3 Indicator Definitions and Formulas

Indicator	Formula / Definition
Elderly Ratio	$(\text{Age } 70-74 + 75-79 + 80-84 + 85+) / \text{Total Population}$
Health Disability	$(T12_1_T + T12_3_BT + T12_3_VBT) / \text{Total Population}$
No-Car Household Ratio	$T15_1_NC / \text{Total Households}$
Economic Vulnerability	$(\text{Long-Term Unemployed} + \text{Permanent Sickness/Disability}) / \text{Economically Relevant Population}$
Normalisation	$X_norm = (X - X_min) / (X_max - X_min)$
Population Vulnerability	$0.30 \text{ Elderly_Norm} + 0.30 \text{ Health_Norm} + 0.20 \text{ NoCar_Norm} + 0.20 \text{ Econ_Norm}$
CAI	$0.50 \text{ Pedestrian Accessibility} + 0.50 \text{ Service Availability}$
CAI_norm	$(CAI - CAI_min) / (CAI_max - CAI_min)$
Access Deficit	$1 - CAI_norm$
VUL_norm	$(\text{Population Vulnerability} - PV_min) / (PV_max - PV_min)$
Hotspot Score	$0.40 \text{ Access Deficit} + 0.60 \text{ VUL_norm}$

Weighting Architecture

Model	Component	Weight
Population Vulnerability	Elderly_Norm	30%
Population Vulnerability	Health_Norm	30%
Population Vulnerability	NoCar_Norm	20%
Population Vulnerability	Econ_Norm	20%
CAI	Pedestrian Accessibility	50%
CAI	Service Availability	50%
Hotspot Score	Access Deficit	40%
Hotspot Score	VUL_norm	60%

Interpretation: The weighting structure is an analytical assumption. Different weights could produce different spatial results and should be examined through sensitivity testing.

Detailed Component Descriptions

Elderly Ratio: The proportion of the total population aged 70 years and over within each Small Area. Higher values indicate a greater concentration of older residents and therefore potentially greater dependence on accessible public transport.

Health and Disability Vulnerability: A composite indicator representing the relative concentration of selected Census health and disability characteristics within each Small Area. Because component categories may overlap at individual levels, it must not be interpreted as a raw percentage of disabled persons.

No-Car Household Ratio: The proportion of households within each Small Area recorded as having no motor car. Measures household vehicle availability as a proxy for potential transport dependence.

Economic Vulnerability: Selected labour-market and economic-status characteristics (long-term unemployed plus permanently sick/disabled) divided by the economically relevant population. Retired persons are treated separately under demographic indicators.

Pedestrian Accessibility: Network-based walking access score from ArcGIS Network Analyst using 5 km/h walking speed across four bands:

Walking band	Accessibility score	Approx. distance at 5 km/h
0–5 minutes	1.0	Up to ~417 m
5–10 minutes	0.6	~417–833 m
10–15 minutes	0.3	~833–1,250 m
>15 minutes	0.0	Greater than ~1,250 m

Composite Accessibility Index (CAI) combines Pedestrian Accessibility (50%) and GTFS Service Availability (50%).

Access Deficit is calculated as $1 - \text{CAI_norm}$, identifying relative access deficiency on a 0 to 1 scale.

Hotspot Score integrates Access Deficit (40%) and VUL_norm (60%). The five resulting planning classes are defined as follows:

Class	Planning Interpretation / Threshold
Priority / Critical Hotspot	Very low accessibility combined with high population vulnerability.
High Concern / High Priority	Low accessibility combined with elevated population vulnerability.
Concern / Moderate	Intermediate Hotspot Score.
Emerging / Low Priority	Lower Hotspot Score where accessibility deficit and/or vulnerability are comparatively limited.
Neutral / Well Served	Lowest Hotspot Score, indicating comparatively stronger accessibility and/or lower relative vulnerability.

3.4 Study Area Context and Census Variables

Census Variable Field	Description	Role in Analysis
T1_1AGE70_74T to GE_85T	Population aged 70+	Included in Elderly Ratio and Population Vulnerability
T1_1AGETT	Total population	Denominator for population-based indicators
T4_5RP / T8_1_RT	Retired persons	Contextual demographic indicator; not directly included in PV formula
T8_1_LTUT	Long-term unemployed	Included in Economic Vulnerability
T8_1_UTWSDT	Unable to work due to sickness/disability	Included in Economic Vulnerability
T12_1_T	Persons with a disability	Included in Health and Disability Vulnerability
T12_3_BT	Bad general health	Included in Health and Disability Vulnerability
T12_3_VBT	Very bad general health	Included in Health and Disability Vulnerability
T15_1_NC	Households without a car	Included in No-Car Household Ratio and Population Vulnerability

Census Variable Field	Description	Role in Analysis
T11_1_BUT	Travelling by bus/minibus	Contextual observed travel-behaviour indicator
T11_2_T1-T8	Departure time for work/school	Contextual commuting-pattern indicator

3.5 Master Map Series

Map ID	Master Title and Subtitle	Key Cartographic Message & Finding
Map 01.00	Base Context & Transit Infrastructure Bandon Built-Up Areas (BUA) and Network Baseline	Establishes spatial boundaries, arterial routes, and the distribution of active bus stops.
Map 01.01	Physical Pedestrian Catchments Network-Based Isochronal Walkability Analysis	Highlights 5, 10, and 15 minute walking catchments at 5 km/h along the road network.
Map 01.02	Pedestrian Network Context Footpath Coverage and Pedestrian Crossing Networks	Provides contextual information on footpaths, crossings and pedestrian-network structure alongside network walking results.
Map 01.03	Headway & Operational Frequency Peak-Period Transit Service Quality	Maps peak-hour bus frequencies and service headways derived from GTFS data.
Map 01.04	Demographic Vulnerability & Demand Spatial Distribution of Relative Population Vulnerability	Displays composite Population Vulnerability across Census Small Areas.
Map 01.05	Composite Accessibility Index (CAI) Combined Pedestrian Accessibility & Service Availability	Combines network Pedestrian Accessibility and GTFS Service Availability (50:50).
Map 01.06	Targeted Intervention Framework Spatial Screening Priorities for Further Investigation	Combines Access Deficit (40%) and VUL_norm (60%) into a five-class spatial screening framework.

3.6 Data Infrastructure and Sources

Data Source	Contribution to the Framework
GTFS Stops, Routes and Trips (NTA)	Public transport locations, routing, and peak timetable
OpenStreetMap	Pedestrian network routing and street topology
CSO Census 2022	Socio-demographic, health, age, and car-ownership indicators
Small Areas – Tailte Éireann / CSO	Primary spatial reporting geography

3.7 Production Stack

- ArcGIS Pro / Network Analyst supports network analysis; PostGIS supports spatial data processing and lin-eage.
- GTFS is used for scheduled service frequency and headway.
- OpenStreetMap provides pedestrian-network context and routing.
- Census 2022 Small Areas provide the population-vulnerability evidence base.
- Tailte Éireann / CSO geography supports the reporting framework.

4. Results and Spatial Interpretation

The map series provides a sequential assessment of public transport accessibility and its population context in Bandon. The principal spatial pattern is stronger network accessibility and scheduled service availability around the town centre, with weaker conditions occurring in several peripheral residential areas.

The maps should be read sequentially. Maps 01.00–01.03 describe the transport network, network walking accessibility, pedestrian-network context and scheduled service availability. Map 01.04 describes the spatial distribution of the Population Vulnerability Index. Map 01.05 combines Pedestrian Accessibility and scheduled Service Availability into the Composite Accessibility Index (CAI). Map 01.06 then combines the resulting Access Deficit with Population Vulnerability to identify accessibility–vulnerability hotspots.

4.1 Map 01.00 – Base Context and Transit Infrastructure

Analysis covers 34 intersecting Small Areas (Pop: 9,863; BUA boundary Pop: 8,196).

This map establishes the spatial baseline for the analysis. It locates the study boundary, built-up settlement, principal road network and active public transport stops used in the subsequent accessibility assessment.

KEY SPATIAL FINDING

The public transport network is concentrated around the established town settlement, creating the spatial baseline for the core–periphery pattern developed through the remaining analysis.

4.2 Map 01.01 – Physical Pedestrian Catchments

Measures access through the pedestrian network using 0–5, 5–10 and 10–15 minute walking bands at 5 km/h.

KEY SPATIAL FINDING

12 Small Areas (35.3%) are within 10 minutes walk of a principal stop, while 22 Small Areas (64.7%) require >10 minutes. 11 Small Areas (32.4%) fall completely outside the 15-minute walking threshold (>1,250 m).

4.3 Map 01.02 – Pedestrian Network Context

Examines pedestrian connections, displaying footpath coverage and crossing locations alongside network walking catchments.

KEY SPATIAL FINDING

Network walking time remains the primary measure of pedestrian accessibility. Footpaths, crossings and other features provide contextual evidence rather than an independently weighted infrastructure-quality score.

4.4 Map 01.03 – Headway and Operational Frequency

Adds the temporal dimension through GTFS scheduled peak-period bus frequency and headway.

KEY SPATIAL FINDING

The central corridor generally has stronger scheduled service availability. Frequency and headway are timetable-based measures, not direct measures of observed waiting time or operational reliability.

4.5 Map 01.04 – Population Vulnerability

Displays the Population Vulnerability Index based on four Census 2022 Small Area domains.

KEY SPATIAL FINDING

- Elderly Ratio (> 70): Spans 0.7% to 20.5% across Small Areas.
- Health and Disability: Ranges from 13.3% to 40.2%
- No-Car Households: Ranges from 0% to 16.5%
- Economic Vulnerability: Ranges from 2.5% to 29.5%
- High Vulnerability Concentration: 6 Small Areas (17.6%) score > 0.75 in normalised vulnerability (VUL_norm)

4.6 Map 01.05 – Composite Accessibility Index (CAI)

Combines network Pedestrian Accessibility with GTFS Service Availability (50:50).

KEY SPATIAL FINDING

8 Small Areas (23.5%) fall into the bottom quartile (< 0.25), demonstrating compounding deficits in walking distance and peak service availability.

4.7 Map 01.06 – Accessibility–Vulnerability Hotspot Framework

Integrates Access Deficit (40%) and VUL_norm (60%) into a five-class screening map.

KEY SPATIAL FINDING

- Critical Coincidence: Exactly 4 Small Areas exhibit both an Access Deficit > 0.75 and VUL_norm > 0.75
- Hotspot Score Range: The composite Hotspot Scores for these 4 areas fall between 0.856 and 0.947, identifying peripheral areas like Gully, Cloughmacsimon, and Kilbrogan as top priorities for further site investigation.

4.8 Cross-Map Synthesis

Across the seven-map sequence, the evidence progresses from the physical transport network baseline to network walking accessibility, pedestrian-network context, scheduled service availability, population vulnerability, integrated transport accessibility and final accessibility–vulnerability hotspots.

- Maps 01.00–01.03 establish the transport network, walking accessibility, pedestrian context and scheduled service availability.
- Map 01.04 adds the population-vulnerability dimension.
- Map 01.05 integrates pedestrian accessibility and scheduled service availability through the CAI.
- Map 01.06 introduces population vulnerability through the final hotspot model.
- Areas of weaker accessibility are not necessarily identical to areas with higher population vulnerability; the hotspot stage identifies their spatial overlap.

4.9 Interpretation of Small Area Indicators

The quantitative analysis is based on 34 CSO Small Areas intersecting the Bandon BUA study area. Because the Small Area boundaries and the BUA boundary are not coincident, some of these statistical units extend beyond the study boundary. The indicator values therefore describe the demographic and accessibility characteristics of the source Small Areas, while the spatial analysis identifies the portions of the study area affected by these conditions.

The aggregate population of the 34 source Small Areas is 9,863, whereas the population attributable to the Bandon BUA study boundary is 8,196. The latter is used throughout this report when describing the population of the study area. The former is used only when describing the underlying Small Area dataset.

4.10 Quantitative Indicator Results Across the 34 Small Areas

Pedestrian accessibility shows substantial variation across the study area. Of the 34 Small Areas, 12 (35.3%) have an accessibility value corresponding to a walking time of 10 minutes or less, while 22 (64.7%) fall within the 10–15 minute or greater-than-15-minute categories. Eleven Small Areas (32.4%) fall into the greater-than-15-minute category.

Eight of the 34 Small Areas (23.5%) have a normalised Composite Accessibility Index of 0.25 or below. The normalised vulnerability measure identifies six Small Areas (17.6%) with values of 0.75 or above. Four Small Areas have both an Access Deficit of at least 0.75 and a normalised Population Vulnerability of at least 0.75, representing top-priority screening hotspots.

5. Evaluation of Research Questions

Evidence dimension	What it contributes
Network walking access	Physical accessibility to principal stops
Pedestrian-network context	Context about footpaths, crossings and network structure
GTFS frequency / headway	Scheduled service availability
Population Vulnerability	Relative concentration of selected demographic, health, economic-status and household characteristics
CAI	Integrated transport-system accessibility
Hotspot Score	Overlap between access deficit and relative population vulnerability

5.1 Research Question One — Spatial Accessibility for Cork City Travel

Public transport accessibility for travel between Bandon and Cork City is spatially differentiated across Bandon. Residents in the central accessibility corridor generally have shorter network walking journeys to principal stops and stronger scheduled peak-period service availability. Several peripheral residential areas have longer walking access and/or weaker scheduled service availability.

Spatial zone	Assessment	Key factors
Central Corridor	Generally stronger conditions	Shorter network walking access; stronger scheduled peak-period availability
Intermediate Areas	Variable / moderate	Variation in walking access and scheduled service availability
Peripheral Areas	Generally weaker conditions	Longer network walking journeys and/or lower scheduled service availability

5.2 Research Question Two — Identifying Potentially Vulnerable Areas and Groups

Affected Group	Primary Barrier	Planning Need
Older Adults (70+)	Potentially greater sensitivity to walking distance and accessibility barriers	Examine stop location, pedestrian routes and accessibility conditions
Disability and poor health	Potentially greater sensitivity to physical barriers and longer access journeys	Examine footpaths, crossings, gradients and accessible vehicle provision
Long-term unemployment / economic status	Potential exposure to transport constraints associated with employment	Examine affordability, service timing and access to employment destinations
Households without a car	Lower household access to private motor vehicles	Examine public transport, walking and other available mobility options

5.3 Spatial Typology of Bandon

Area type	Spatial expression	Interpretation
01 — Higher Access / Lower Vulnerability	Town core and central accessibility corridor	Generally stronger transport-system accessibility and comparatively lower relative vulnerability.
02 — Intermediate Access / Mixed Vulnerability	Transitional residential areas	Variable accessibility and mixed vulnerability conditions; location-specific assessment required.
03 — Lower Access / Higher Vulnerability	Several peripheral residential areas	Weaker relative transport accessibility overlaps with higher vulnerability scores.

5.4 Evidence-to-Decision Chain

Evidence	Decision-support role
Map 01.01	Identify locations with longer network walking access.
Map 01.02	Contextualise pedestrian routes, footpaths and crossings.
Map 01.03	Identify locations for operational service-frequency assessment.
Map 01.05	Provide integrated transport-system accessibility evidence.
Map 01.06	Identify candidate locations for detailed investigation.

6. Policy and Planning Implications

PLANNING INTERPRETATION

Critical Hotspot and High Priority areas should be treated as candidate locations for detailed planning and feasibility assessment, rather than automatic investment decisions.

Planning theme	Evidence-led implication
Priority intervention areas	Use Map 01.06 to identify candidate locations for detailed assessment.
Pedestrian infrastructure	Use Maps 01.01 and 01.02 to examine access routes, footpath continuity and crossings.
Service frequency	Use Map 01.03 where walking access is relatively strong but scheduled service availability is weaker.
Flexible mobility	Assess DRT, feeder services or other flexible options where fixed-route catchments are weak.
Monitoring	Repeat the GIS workflow as GTFS schedules, population and pedestrian infrastructure evolve.

6.1 Priority Intervention Areas

Critical Hotspot and High Priority areas in Map 01.06 should be treated as candidate locations for detailed planning and feasibility assessment, rather than as automatic investment decisions.

6.2 Pedestrian Infrastructure Improvements

- Investigate footpath continuity along access corridors.
- Assess pedestrian crossings and route safety.
- Investigate physical barriers where site-level evidence confirms that they constrain practical access.
- Prioritise site verification before selecting an infrastructure response.

6.3 Service Frequency Optimisation

- Review scheduled service frequency and headway patterns.
- Consider timetable structure in relation to observed demand.
- Assess operating constraints and network-wide service objectives before implementation.

6.4 Flexible Mobility and Demand-Responsive Transit (DRT)

Peripheral areas outside or weakly served by fixed-route catchments may warrant assessment of Demand-Responsive Transit (DRT), local feeder services or other flexible mobility options.

6.5 Accessibility Monitoring

Option	What the current study supports	What requires further work
Demand-Responsive Transit	Peripheral areas outside or weakly served by fixed-route catchments may warrant assessment.	Demand, operating requirements, cost, accessibility and network integration.
Local feeder services	Potential response to weak peripheral fixed-route access.	Demand and operational feasibility.
Accessibility monitoring	Workflow can be rerun as inputs change.	Repeatable production metadata and updated source extracts.

6.6 Recommendations for Local Authorities and NTA

- **Targeted Spatial Screening:** Adopt Map 01.06 as a baseline decision-support layer for directing active travel, pedestrian infrastructure, and public transport investments toward identified Critical Hotspots.
- **Peripheral Infrastructure Audits:** Prioritise immediate site-level pedestrian surveys along key arterial routes connecting peripheral residential areas (Gully, Cloughmacsimon, Kilbrogan) to central bus stops.
- **Service Frequency Optimisation:** Re-evaluate peak and off-peak headways on regional routes connecting Bandon to Cork City.
- **Flexible and Demand-Responsive Transit:** Conduct feasibility assessments for micro-transit or Demand-Responsive Transit options targeting outer peripheral neighbourhoods beyond 15-minute walking catchments.

7. Limitations and Future Research

7.1 Limitations

Limitation	Implication
Walking assumptions	5 km/h standard; does not model individual ability, gradients, weather, safety or route preference.
Network representation	Mapped stops and routes depend on available datasets; local conditions require verification.
GTFS schedules	Scheduled timetable information does not represent real-time delay, cancellation or observed waiting time.
Census geography	2022 Small Areas were redrawn; comparison with earlier releases requires geographic harmonisation.
Population vulnerability	Model-derived spatial proxy, not individual dependence or observed demand.
Relative normalisation	Scores are relative within the Bandon analytical population.
Demand dynamics	Potential spatial access to supply, not realised individual trip behaviour.
Composite weighting	50:50 CAI and 40:60 hotspot weights are analytical assumptions.
Classification thresholds	Final classes depend on thresholds that should be recorded explicitly.

7.2 Future Research Priorities

Priority	Future development
Real-time data	Integrate real-time data to evaluate reliability and delay impacts.
Temporal expansion	Evaluate off-peak, evening and weekend accessibility.
Schools and healthcare	Measure access to essential destinations.
Active travel	Integrate walking and cycling infrastructure quality, safety and connectivity.
Quality of Life Index	Develop transport accessibility as a spatial quality-of-life dimension within the Ireland Small Area QoLI framework.
Regional replication	Replicate the workflow across comparable Irish regional towns such as Clonakilty, Skibbereen, Bantry, Midleton and Mallow.

8. Conclusion

- The principal result is a clear spatial differentiation between the more strongly connected central area and several peripheral residential areas with weaker accessibility conditions.
- Population vulnerability is not distributed identically to transport accessibility, supporting separate treatment of the two dimensions before identifying overlap.
- The resulting framework provides a structured basis for identifying areas that warrant further pedestrian, service or flexible-transport investigation.

8.1 Summary of Findings

The 34-Small-Area analysis demonstrates substantial spatial variation in public transport accessibility and population vulnerability across Bandon. Network-based pedestrian accessibility is uneven: 12 of the 34 Small Areas fall within a 10-minute walking threshold to a principal public transport stop, while 22 require more than 10 minutes and 11 are beyond the 15-minute threshold.

When transport-system accessibility is combined through the Composite Accessibility Index, eight Small Areas fall within the lower quarter of the observed CAI_norm distribution. The subsequent Access Deficit and Population Vulnerability analysis identifies four Small Areas where both dimensions are simultaneously high (Hotspot Scores between 0.856 and 0.947).

8.2 Research Contribution

The principal contribution is methodological: the study demonstrates a structured GIS workflow for assessing public transport accessibility in a smaller Irish town using spatial network data, GTFS schedules and Census 2022 Small Area information.

8.3 Final Reflection

The analysis demonstrates the value of moving beyond simple straight-line distance when examining public transport accessibility. In Bandon, accessibility conditions vary according to network walking access to principal stops and scheduled service availability, while population characteristics provide an additional dimension for assessing potential exposure to transport disadvantage.

References

- Accessibility Theory: Hansen (1959); Geurs & van Wee (2004)
- Transport Equity: Lucas (2012); Martens (2016)
- GIS Accessibility: ESRI Network Analyst Literature & OpenStreetMap Routing Protocols
- Irish Transport Policy: National Transport Authority (NTA); Department of Transport
- Census & Spatial Analysis: Central Statistics Office (CSO) Census 2022

Appendix A. Data Sources and Reproducibility Register

This appendix records the principal datasets used in the SEL-002 analytical workflow. To support independent reproduction of the analysis, the final production record should identify the dataset version, extraction date, source, spatial reference, and processing date for each dataset.

Dataset	Version / date	Purpose	Production record
CSO Census 2022 Small Areas	2022	Population and socio-demographic indicators	Retain source release and extraction record.
Bus Éireann / NTA GTFS	GTFS feed	Stops, routes and stop times	Record exact feed/extraction date for archive.
OpenStreetMap Extract	2026	Walking street network and local context	Record exact extract date.
Administrative Boundaries	Tailte Éireann / CSO	Study and reporting geography	Retain layer/version metadata.
SEL PostGIS Database	Version 1.0	Processed accessibility and vulnerability layers	Record schema/table lineage and processing date.

Appendix B. Map Production Register

Map	Title	Subtitle
01.00	Base Context & Transit Infrastructure	Bandon Built-Up Areas (BUA) and Network Baseline
01.01	Physical Pedestrian Catchments	Network-Based Isochronal Walkability Analysis
01.02	Pedestrian Network Context	Footpath Coverage and Pedestrian Crossing Networks
01.03	Headway & Operational Frequency	Peak-Period Transit Service Quality
01.04	Demographic Vulnerability & Demand	Spatial Distribution of Relative Population Vulnerability
01.05	Composite Accessibility Index (CAI)	Combined Pedestrian Accessibility and Scheduled Service Availability
01.06	Accessibility – Vulnerability Hotspot Framework	Spatial Screening Priorities for Further Investigation

Appendix C. Executive Summary

SEL-002: Bandon Public Transport Accessibility

9,863 AGGREGATE POPULATION 34 Census Small Areas in Bandon Built up Area (BUA)	64.7% WALK ACCESS DEFICIT 22 of 34 areas require >10 min walk to transit	32.4% OUTSIDE CATCHMENT 11 areas sit completely outside 15-min walk zone
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Core Findings & Value Proposition

This independent geospatial analysis establishes a spatial screening framework across 34 Census Small Areas intersecting the Bandon Built-Up Area (BUA study population: 8,196; aggregate Small Area population: 9,863). The research decouples transport network performance from population demographic need to identify systemic deficits and targeted investment opportunities.

- Centralised Network Bias:** 64.7% of evaluated Small Areas (22 of 34) require network walking times exceeding 10 minutes to reach principal transit stops, with 32.4% (11 Small Areas) falling completely outside the 15-minute network walking catchment (>1,250 m).
- Strategic Hotspot Isolation:** By combining an Access Deficit measure (40% weight) with a normalized Population Vulnerability Index (60% weight—encompassing older age, health/disability, car availability, and economic status), the model isolated 4 critical hotspot Small Areas. These areas (including Gully, Clough-macsimon, and Kilbrogan) exhibit composite Hotspot Scores between 0.856 and 0.947, where severe transit deficits (Access Deficit > 0.75) directly overlap with high social vulnerability (VUL_norm > 0.75).

Hotspot Evaluation Model Architecture

COMPOSITE HOTSPOT SCORE (100%)	
ACCESS DEFICIT (40% Weight)	POPULATION VULNERABILITY (60% Weight)
Pedestrian Walkability <i>Sub-weight: 50% Weight</i> GTFS Scheduled Peak Frequency <i>Sub-weight: 50% Weight</i>	Elderly Population Rate <i>Sub-weight: 30% Weight</i> Health & Disability Ratio <i>Sub-weight: 30% Weight</i> No-Car Household Ratio <i>Sub-weight: 20% Weight</i> Economic Vulnerability <i>Sub-weight: 20% Weight</i>

Critical Priority Hotspot Small Areas

Small Area / Location	Access Deficit	Vulnerability Index	Composite Score
Gully Peripheral Zone	0.892 (Severe)	0.884 (High)	0.947
Cloughmacsimon Area	0.810 (High)	0.865 (High)	0.895
Kilbrogan Outer Sector	0.785 (High)	0.820 (High)	0.868
Bandon West Corridor	0.760 (High)	0.790 (Moderate)	0.856

Capital Allocation & Operational Recommendations

Targeted Active Travel Infrastructure Prioritise footpath connectivity, pedestrian crossings, and gradient improvements along arterial routes connecting outer residential areas (Gully, Cloughmacsimon, Kilbrogan) to central transit nodes.
Service Frequency Optimisation Re-evaluate GTFS scheduled peak-period headways where pedestrian network access is high but service frequency remains low.
Demand-Responsive Transit (DRT) Conduct feasibility assessments for micro-transit or DRT solutions in outer peripheral zones situated beyond the 15-minute walking catchment, rather than extending costly fixed-route infrastructure.

DOCUMENT CONTROL

Project: SEL-002 – Bandon Public Transport Accessibility Analysis

Status: Final Revised Report (Updated Master Map Titles and Strategic Synthesis)

Date: September 2026

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